

Hongrui (Jack) Zhang

Computer Science Ph.D. Researcher | Data Systems and GPU Computing

Irvine, California | 302-898-6341 | jackzhang010603@gmail.com

LinkedIn | hongruizhang.com

EDUCATION

Ph.D. in Computer Science, University of California, Riverside

Sep 2024 – Expected Jun 2028

Advisor: Hung-Wei Tseng

B.S. in Computer Engineering, University of California, San Diego

Sep 2019 – Sep 2023

RESEARCH EXPERIENCE

ESCAL Laboratory, University of California, Riverside | Advisor: Hung-Wei Tseng

GPU-Accelerated Database Systems and Matrix-Native Query Processing

Jun 2026 – Present

Python, PyTorch, SQL, CUDA, cuSPARSE | Query execution and experimental evaluation

- Developed and correctness-validated a matrix-native algorithm for TPC-H Q3, reformulating joins, group-by, and aggregation using sparse matrix and tensor operations.
- Optimized TCUDB-style sparse joins using direct one-hot CSR construction, reducing Q4 join latency from 40.5 ms to 11.0 ms (3.7×) over generic COO construction at TPC-H SF10.
- Implemented and SQL-validated PyTorch execution plans for all 22 TPC-H queries, profiling GPU bottlenecks on an RTX 5090 to guide hardware-aware query processing.

RTSpMSpM: Harnessing Ray Tracing for Efficient Sparse Matrix Computations

Sep 2023 – Sep 2025

ISCA '25 | First author and conference presenter | OptiX, CUDA, C++, Python

- Designed an algorithm that maps sparse matrix-matrix multiplication to ray-tracing operations, enabling NVIDIA RT Cores to accelerate non-graphics sparse computation.
- Developed a library that realized the proposed algorithm using OptiX and CUDA to achieve 1.85× speedup over cuSPARSE on NVIDIA's RT Cores.
- Proposed RT core architectural enhancements, including fused intersection-multiplication and hardware row accumulation, achieving 3.06× simulated speedup over cuSPARSE with only 0.2% estimated area overhead.
- Integrated the library into a graph neural network training framework using CUDA and Python, achieving more than 3× speedup on NVIDIA RTX 4090.

SELECTED PUBLICATION

Hongrui Zhang, Yunan Zhang, and Hung-Wei Tseng. "RTSpMSpM: Harnessing Ray Tracing for Efficient Sparse Matrix Computations." *52nd International Symposium on Computer Architecture (ISCA '25)*, 2025. [DOI]

TECHNICAL SKILLS

Languages used in research: Python, C++, SQL, CUDA C++, C, Bash.

Data systems: Relational and tensor algebra, query execution, joins and aggregation, TPC-H, MonetDB, TQP-style tensor plans, TCUDB reproduction, sparse COO representations.

Research software: PyTorch, CUDA, NVIDIA OptiX, cuSPARSE, NumPy, XGBoost.

Experimental methods and tools: SQL correctness validation, controlled benchmarks, end-to-end evaluation, Nsight Systems, Nsight Compute, PyTorch Profiler, roofline analysis, NVBit memory tracing, CUDA events.

Development tools: Linux/Ubuntu, Git, Docker, gdb, vim.